Develop the Core Functionalities

Workout Plan Generation Project

Story Duration: 1 hour 30 minutes

1. Workout Plan Generation

• **Function:** generate_workout_gemini()

• **File:** gemini_generator.py

• Implements the core logic to generate a personalized 7-day workout plan using user inputs such as goal (e.g., muscle gain, weight loss) and intensity (low, medium, high).

• The prompt is sent to the Gemini 1.5 Pro model, which returns a structured, day-wise plan including:

• Warm-up (5–10 mins)

• Main workout (exercise details, sets, reps)

• Cooldown or recovery tip

• The output is returned to routes.py and rendered on the result.html page using blocks for formatting.

2. Nutrition Tip Generation

• **Function:** generate_nutrition_tip_with_flash()

• **File:** gemini_flash_generator.py

• This function generates a concise, practical nutrition or recovery tip tailored to the user's selected goal.

• It utilizes the Gemini Flash model to deliver a fast, helpful response such as hydration advice, protein suggestions, or general recovery best practices.

• This tip is displayed beneath the workout plan in result.html.

3. Feedback-Based Plan Updating

• **Function:** update_workout_plan()

• **File:** updated_plan.py

• This function enables the user to revise their original plan by submitting feedback (e.g., "Add yoga", "Include more cardio").

• The original plan and feedback are passed to Gemini 1.5 Pro, which returns a modified plan with the suggested updates.

• The updated plan is then stored and displayed alongside the new nutrition tip.

4. User & Plan Storage

• **Functions:** save_user(), save_plan(), update_plan(), get_original_plan(), get_user()

• **File:** database.py

• All user information (ID, name, age, weight, goal, intensity) and AI-generated plans are stored in a SQLite database via SQLAlchemy ORM.

• Feedback-based updates are saved as a separate field to preserve both versions (original and updated).

5. Admin View of All Users

• **Route:** /view-all-users

• **Template:** all_users.html

• Displays a list of all users and their respective plans.

• Rendered with Jinja2 using a {% for user in users %} loop.

• It allows administrators to see original and updated plans clearly in a structured table format.

System Flow

• User enters personal details, workout goal, and intensity.

• Gemini 1.5 Pro generates the personalized 7-day workout plan.

• Gemini Flash generates a concise nutrition or recovery tip.

• The plan and tip are displayed on result.html.

• User can submit feedback to modify the workout plan.

• The updated plan is stored separately while preserving the original plan.

• Administrators can review all users and their plans through /view-all-users.